SruthiScribe

Sing it. Read it back as svaras.

A browser-based Carnatic transcription engine and an open, machine-readable library of the repertoire.

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THE PROBLEM

Carnatic music is taught by ear, and stays there

A student cannot see what they just sang

Correction happens in the lesson, from the teacher's ear. Between lessons there is no feedback at all.

Notation exists, but not where you need it

Printed books, PDFs and scanned archives — none of it queryable, none of it aligned to what you are singing.

The reference corpus is scattered and closed

Ragam definitions on one site, lyrics on another, notation in a 1904 book. Little of it is machine-readable.

What a learner actually has today

A tanpura app for the drone.
A tuner that says whether one note is sharp.
A PDF of notation they cannot search.
A recording of their own practice they will never listen back to.

Nothing that turns the singing into notation.

THE NEED

Two halves of one gap

1 · Hear yourself as notation

Sing a phrase; get back svaras with sthayi, gamaka and avartana bars, laid out the way the tradition prints it.

Practise between lessons with something that answers.

2 · One library, openly licensed

Compositions, ragams, talams and notation in one place, searchable, with the source and licence of every row recorded.

So a reading can be matched against the repertoire.

The two halves feed each other: the library gives the decoder its ragam and tala; the decoder gives the library new readings.

FUNCTIONALITY

What it does today

Record or upload, and read it back

Microphone or file, in the browser. Svaras with octave, duration, gamaka class and cents deviation.

Tala-aware layout

All seven suladi sapta talas × five jatis, nadai 1x–8x, avartana and anga bar lines, sahitya under the svaras.

Fix what it got wrong

Edit any svara, mark the eduppu, regroup speeds — the decode is a starting point, not a verdict.

Export

Print-quality PDF laid out as the tradition prints it, JSON, or contribute the reading to the library.

A library of 598 compositions

42 with complete notation, searchable by ragam, tala, composer, deity and how complete the notation is.

939 ragams

All 72 melakartas and their janyas, with arohana/avarohana; 114 wired into the decoder as constraints.

Provenance on every row

Source, licence and page, so a notation can be checked against the book it came from.

An AI teacher, optional

Reads the transcription and answers questions about it. Never part of producing the transcription.

TECHNICAL ARCHITECTURE

A signal-processing pipeline, entirely in the browser

Audio	YIN	Voicing	Tonic	Viterbi	Notation
mic or file 16 kHz mono	pitch track 50 ms window	energy + confidence 90 ms min. note	pitch histogram auto-sruthi	ragam-constrained svara decode	tala grid PDF / JSON

No server, no model, no key

The engine is 44 kB of dependency-free JavaScript. Transcription works offline; nothing is uploaded.

The ragam is the decoder's state space

Viterbi over that ragam's svarasthanas, penalising switches — so the decode cannot invent a note outside the ragam.

Postgres + REST for the library

Supabase; the page talks to it directly. Notation stored as JSON with the tala and akshara grid.

The harness scores the shipping code

tools/eval.js extracts the engine from the page itself and scores it against public datasets — the two cannot drift.

MEASURED, NOT CLAIMED

Accuracy on studio recordings

91.7%	82.5%	2.5%	3.2¢
svara accuracy (label + octave)	raw pitch accuracy	octave error rate	tonic (sruthi) error

Scored on 33 items of Sanidha (Georgia Tech), studio multitrack, against the dataset's own pitch annotations.

How the measurement is kept honest

The harness runs the engine that ships, not a copy. Records whose published annotation disagrees with their own audio are detected and excluded rather than averaged in — three of Sanidha's are annotated an octave high. Every engine change is confirmed on a held-out split before it lands.

THE LIBRARY

Rescuing a 1904 book by machine

Subbarama Dikshitar's Sangita Sampradaya Pradarsini (1904) is the only public-domain source of complete Carnatic notation — and it has never been machine-readable.

598	**42**	**939**	**4,695**
compositions	with complete notation	ragams	svaras extracted from SSP

The extractor reads the book's own notation scheme from its page iii — capitals are two aksharas, underlines halve, dots mark sthayi — and verifies every section against the tala the book prints for it. 26 kirtanas parse clean end to end; 12 of them were confirmed independently against a separate catalogue, agreeing on both title and ragam. Nothing that fails verification is loaded.

LIMITATIONS

What it cannot do yet

Ragam identification is weak — 15% top-1

A pitch histogram cannot separate ragams that share a svara set and differ by phrase. The user still has to name the ragam.

Gamaka classification is coarse

Five machine labels, not the dasavidha taxonomy. It says 'kampita', not which kampita.

Voicing false alarm is 32%

It still calls notes in the gaps. The weakest number in the engine, and known.

One voice, no ensemble

Trained on and scored against solo vocal. Accompaniment in the room degrades it.

The library is mostly metadata

433 of 598 rows have no notation. Complete notation is not openly licensed at scale; that is a licensing wall, not an engineering one.

The SSP extractor verifies 44% of sections

Errors are per-glyph now — underline geometry, tight spacing. Each further point is slow work.

No Tyagaraja notation

SSP covers Dikshitar. The Walajapet manuscripts are scans with no text layer.

Not a substitute for a teacher

It measures pitch. It does not hear bhava, and it should not be sold as if it did.

LANDSCAPE

What is genuinely new here

Already exists

- Tuners that name the svara you are holding — Carnatic Tuner, Shruti.
- Practice apps with pitch feedback and lesson tracks — Carnatic Singer, Riyaz.
- AI accompaniment and raga detection — NaadSadhana, an Apple Design Award winner with 410 ragas.
- Notation typing tools — Haathi Carnatic Editor.
- Research corpora — CompMusic, Saraga, Sanidha.

Less crowded

- Continuous transcription of a phrase into tala-aligned notation, not just the current note.
- Runs entirely in the browser — no install, no account, works offline.
- Published accuracy against public datasets, with the harness in the repo.
- **A machine-readable SSP corpus with provenance — as far as we know, the first.**

Honest reading: the transcription feature is differentiated, not unique. The extracted SSP corpus is the genuinely novel contribution.

WHERE IT GOES

Next

- Raise the SSP yield — every point converts directly into complete works.
- Cut the voicing false alarm; it is the weakest measured number.
- Phrase-level ragam identification, where the published work actually lives.
- Attach openly-licensed audio to notated works, so a student can hear and read together.

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